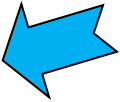


Data Mining Methodologies and Systems

- ❑ Structuring Unstructured Data for Knowledge Mining: A Data-Driven Approach
- ❑ Data Augmentation 
- ❑ From Correlation to Causality
- ❑ Network as a Context
- ❑ Auto-ML: Methods and Systems

Data Augmentation Details

□ Why do we need data augmentation?

- A high-performing data mining model often requires a great amount of labelled data samples for training
- However, for many domains, we might be only provided by a limited number of labelled data
- Training a data mining model with scarce training data, which could lead to the overfitting issue

□ How

- Basic Methods: Flipping, rotation, equalization and so on
- Augmentation Policy Learning: Combine multiple basic augmentations
- GANs-Based Methods: Training a GAN model to generate new data
- Adversarial Training: Mixing the trickily-perturbed samples into the training samples